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I&D (Informatica + Snowflake) LoT Course Structure

I&D (Informatica + Snowflake) LoT provides exposure to a band of data warehousing technologies. It focuses on application development for data warehouses. The following table lists the course structure for I&D LoT.

Sr. No.	Course	Duration (In Days)	Remarks
1	Discover(Induction)	-	Online
2	Soft Skills Part 1	1	Soft Skills Part 1
3	Introduction to Software Engineering	0.5	
4	Introduction to software development methodologies (waterfall, agile, and DevOps)	0.5	
5	UNIX	3	
6	RDBMS + Oracle SQL	7	Project Kick Off
7	Soft Skills Part 2	1	Soft Skills Part 2
8	Module 1 Assessment		MCQ
9	Data Warehouse Concepts	1	Sprint 1 Implementation, MCQ
10	Data Modelling for Business Intelligence	1.5	
11	ETL Basics	0.25	
12	Software Testing for BI	0.25	
13	Soft Skills Part 3	1	Soft Skills Part 3
14	Cloud Fundamentals	0.5	Sprint 1 Implementation, MCQ
15	AWS Cloud Basics	2.5	
16	Snowflake	5	
17	Sprint 1 Evaluation	1	Sprint 1 Evaluation
18	Informatica Power Center	8	Sprint 2 Phase 1 Implementation, MCQ
19	Soft Skills Part 4	1	Soft Skills Part 4 (Evaluation)
20	Sprint 2 Phase 1 Evaluation	1	Sprint 2 Phase 1 Evaluation
21	Informatica IICS	9	Sprint 2 Phase 2 Implementation, MCQ
22	Sprint 2 Phase 2 Evaluation	1	Sprint 2 Phase 2 Evaluation
23	L1 Preparation	1	
24	L1 Test	1	MCQ
Total Training Duration		**48**	

I&D Curriculum

Introduction to Software Engineering

Program Duration: 0.5 day
Contents:

- To Understand the following :
  - What is Software Engineering (SE)
  - Common life cycle models
  - Phases in SE
  - Familiarizing Requirements Phase
  - Familiarizing Design Phase
  - Familiarizing Construction Phase
  - Familiarizing Testing and acceptance Phase
  - Review and Configuration Management Process

Introduction to Software Development Methodologies

Program Duration: 0.5 day
Contents:

- To Understand the following:
  - Development Methodologies
    - Waterfall
    - Agile
      - Agile Manifesto
      - Basics of Scrum including various roles & responsibilities, phases (Product Backlog, Sprint planning, Backlog, Sprint), Ceremonies (STUM, Sprint review, retrospective)

UNIX

Program Duration: 3 days.
Contents:

Introduction to UNIX Operating System and Basic UNIX commands

Introduction to UNIX Operating System and Basic UNIX commands
Operating System
Basic UNIX Commands

UNIX File System

UNIX File System
File Types
File Permissions
File Related Commands
Filters
Simple Filters
Advanced Filters

Vi Editor
Vi Editor
Input Mode Commands
Vi Editor – Save & Quit
Cursor Movement Commands

Shell Programming
Shell Variables
Environmental Variables
Shell script Commands
Arithmetic Operations
Command Substitution
Command Line Arguments
Conditional Execution
if Statement Format
Test - String Comparison
The Case Statement
While Statement
Break & Continue Statement
Until Statement
Shell functions
Using arrays

Oracle SQL

Program Duration: 7 days.

Contents:

Introduction to Database
Introduction to DBMS
Characteristics of DBMS
DBMS Models
Relational DBMS
Data Integrity

Normalization & Codd's Rules for "FULLY" Functional System

- First Normal Form
- Second Normal Form
- Third Normal Form
- Relational DBMS
- Data Integrity

Structured Query Language

- Interacting SQL using SQL *Plus
- Using SQL *Plus
- What is SQL?
- Rules for SQL statements
- Standard SQL Statement Groups
- Basic DataTypes
- Rules for naming a Table
- Specifying Integrity Constraints
- DDL Statements: Create, Alter, Drop
- Regular vs Temporary tables

Data Manipulation Language

- Inserting Rows Into a Table
- Deleting Rows from a Table
- Updating Rows in a Table

Database Objects

- Index
- Synonym
- Sequence
- Views

Data Query Language (Select Statement)

- Select Statement
- Distinct Clause
- Comparison, arithmetic & Logical Operators SQL Operators
- The ORDER BY Clause
- Tips and Tricks

Aggregate Functions, Group By and Having Clause

- Aggregate Functions
- The GROUP BY Clause
- HAVING Clause
- ROLLUP Operation
- CUBE Operation
- Tips and Tricks

SQL (Single Row) Functions

- Character Functions
- Number Functions
- Data Conversion Function

- Formats for Date functions
- Date Functions
- Miscellaneous Functions
- Tips and Tricks

Transactions

- Transaction
- Commit Command
- Rollback and Savepoints

Joins and Subqueries

- Inner/Equi Join
- Outer Join
- Self Join
- Subquery
- SUBQUERIES Using Comparison Operators Co-related Subquery
- Exists / Not Exists Operator
- Connect By and Start with clauses
- Tips and Tricks

Set Operations

- The UNION Operator
- The INTERSECT Operator
- The MINUS Operator
- The UNION Operator
- The INTERSECT Operator
- Tips and Tricks

Data Control Language

Introduction to Oracle Architecture

Introduction to Data Dictionary

Analytical Functions

- RANK & Dense Rank
- Max, Min, Sum & Avg
- Row Number
- Lead & Lag
- FIRST_VALUE & LAST_VALUE
- LISTAGG
- WITH CLAUSE

SQL * Plus Reports

- SQL * Plus Reporting
- SQL * Plus Commands

SQL * Loader

- What is SQL * Loader?
- SQL * Loader as a Utility
- SQL * Loader Environment

The Bad File and Discard File
Invoking SQL * Loader
SQL * Loader Examples
Oracle Tools

Data Warehouse Concepts

Program Duration: 1 day.

Contents:

Business Intelligence
- Business Intelligence
- Need for Business Intelligence
- Terms used in BI
- Components of BI

General concept of Data Warehouse
- Data Warehouse
- History of Data Warehousing
- Need for Data Warehouse
- Data Warehouse Architecture
- Data Mining Works with DWH
- Features of Data warehouse
- Data Mart
- Application Areas

Dimensional modeling
- Dimension modeling
- Fact and Dimension tables
- Database schema
- Schema Design for Modeling
- Star
- Snow Flake
- Fact Constellation schema

ETL and Metadata
- ETL process
- Metadata used in ETL
- Metadata in Data Warehousing
- Simple Data warehouse model

Online Analytical Processing (OLAP)

- Online Analytical Processing (OLAP)
- Nature of OLAP analysis
- Types of OLAP
- OLAP Tools
- OLTP and OLAP
- OLAP Functional requirements
- OLAP Fast and Selective
- Operational versus Informational System

Data Mining

- Data mining
- The Knowledge Discovery process
- Need of Data Mining
- Use of Data mining
- Data mining and Business Intelligence
- Types of data used in Data mining
- Data Mining applications
- Data Mining products
- Data Mining market

Best Practices for Building Data Warehouse

- Recipe for a Successful data warehouse
- Data warehouse pitfalls
- Popular BI DW tools and suits
- Trends in BIDW

Data Modeling for Business Intelligence

Program Duration: 1.5 days

Contents:

- Introduction to Data Modeling
  - Importance of data modeling
  - Features of a good data model
  - Who should be involved in data modeling
  - Database design stages and deliverables
  - Classification of information
- Understanding Business Requirements
  - Need of Requirement Analysis
  - Characteristics of a Good Requirement
  - The Data Life cycle
  - Methods of Collecting requirement
  - Business Requirement Specification (BRS)
- Conceptual Model

Define conceptual model
Objectives of conceptual model
Components of Conceptual Model
Types of Modeling
Entity-Relationship (ER) model
Types of Attributes
Join Problems
Steps of dimension modeling
Star Schema
Snowflake Schema
Bill Inmon Vs Ralph Kimball Approach

- Logical Model
  - Define logical model
  - List features of a logical model
  - Transformations required to be done while converting a conceptual model into a Logical model
  - Activities in table specification
  - Activities in column specification
  - Activities in Primary key specification

ETL Basics

Program Duration: 0.25 day
Contents:

- Basic Concepts
  - Data warehouse
  - Data warehousing strategies
  - Data warehouse architecture
  - ETL Meaning
  - Need for ETL
  - ETL Process
  - Operational Considerations
- ETL Process
  - Data extraction
  - Data transformation
  - Data Loading
- Operational Considerations
  - Exceptional Handling
  - Alerts and Notification

Process restart-ability
Job Scheduling and Monitoring

ETL Tools

Leading ETL tool vendors
ETL tool strengths / weaknesses
Choosing the correct ETL tool

Software Testing for BI

Program Duration: 0.25 day
Contents:

- Introduction to Software testing for BI
  - Business requirements
  - BI Project versus BI Program
  - How is BI testing different from traditional code based testing?
  - BI SDLC
- Testing concepts
  - What is testing? Testing – Why? Testing – How?
  - Principles of Testing
  - Test Case and Test Suite
  - Testing scope
  - Test Strategy
  - Verification and Validation
- Types of Testing
  - Static Testing, Dynamic Testing, Automated testing
  - V Model for BI Testing
- Testing for BI
  - Testing document purpose (Test documentation)
  - General BI Testing Principles
  - BI Testing Mission
  - Production Verification Testing
  - Possible Areas of Automation

Cloud Fundamentals

Program Duration: 0.5 days
Contents:

- What is Cloud Computing?
- Cloud Deployment Models
- Key Cloud Concepts
- Cloud Service Models
- Cloud providers and details

AWS Cloud Fundamentals

Program Duration: 2.5 days
Contents:

- AWS Overview
- Various AWS Services
- Global Infrastructure – Regions and Availability Zones
- Understanding Identity Access Management of AWS
- EC2 Instance
- Auto Scaling
- Load Balancing
- EBS (Elastic Block Storage)
- Object Storage
- Amazon Virtual Private Cloud (VPC)
- Relational Database Service (RDS)
- Monitoring Services

Snowflake Basics

Program Duration: 5 days (Including Lab)
Contents:

Snowflake Architecture and Overview

- Snowflake technical overview
- Review three-tiered architecture
- Cloud services
- Virtual warehouse
- Storage

Interfaces and Connectivity

- Using Snowflake web console, worksheet
- Using Snowflake command line, SnowSQL
- Overview of connectors and ecosystem for Snowflake

Developing for Snowflake

- Overview of programmatic interfaces for Snowflake
- A deeper look at a specific programming interface such as Python and Spark

- Stored procedures
- External functions
- User defined functions

Loading New Data Sets

- Ingesting new data into Snowflake tables
- Working with various SQL data types
- Ingestion Snowflake best practices

Data Pipelines

- Continuous data processing using tasks
- Continuous data ingestion using SnowPipe
- Continuous change processing using streams
- Streaming connectors, such as Kafka for continuous ingestion

Data Lakes

- External tables and data lakes
- Partitioning for e_ecient queries over external files
- Materialized views sourced from external tables

Pipeline Transformations and Querying Data

- Review best practices of writing e_ective queries
- Filtering data examples and best practices
- Walk through grouping, rollups, and sorting data and performance considerations
- Usage and pitfalls of joining data
- Using Snowflake's high performing approximation and estimation functions
- CTEs and analytic functions

Query Caching Performance Features

- Result set cache
- Metadata cache
- Query data cache
- Best practices of using caching for performance and cost optimization

Performance monitoring and management of query and ETL workloads

- Query profiling
- Virtual warehouse (compute resource) management
- Optimizing and tuning workloads
- Monitoring functions and cost management

Using Data Clustering Optimization for Advanced Query Performance Tuning

- How to identify appropriate use cases
- Designing clustering keys

- Auto-clustering service
- DML considerations
- Materialized views
- Search optimization service

Test, QA, and Production and Agile Development

- Time travel queries in Snowflake
- Cloning data and environment in Snowflake

Working with Semi-Structured Data

- Data source formats
- Support of native data types
- SQL operations (grouping, sorting and more)
- Built-in functions for traversing, flattening, and nesting of semi-structured data

Snowflake Data Cloud

- Data sharing Overview
- Snowflake Data Exchange and Snowflake Data Marketplace
- Secure views and UDFs

Informatica PC + IICS

Program Duration: 17 days

Informatica PC

Program Duration: 8 days

- Introduction to Informatica Powercenter
  - Informatica PowerCenter Components
  - PowerCenter Domain
  - Integration Service
- Powercenter Designer
  - Concept of a Mapping
  - PowerCenter Designer : Modes
  - Analyze Sources
  - Creating a Target Schema
  - Mapping Designer
- Transformation Overview
  - Source Qualifier Transformation
  - Expression Transformation
  - Aggregator Transformation

- Basic Transformations
    - Lookup Transformation
        - Sharing Cache
        - Persistent Cache
        - Non Persistent Cache
    - Filter Transformation
    - Joiner Transformation
- Working with Flat Files
    - Overview Flat File Properties
- Mapplet
    - Concept of Mapplet
    - Mapplet Input and Output
- Workflow Manager
    - Workflow
    - Workflow Monitor
    - Database Connections
    - Tasks
    - Session Task
    - Debugger
    - Load Manager Process
    - DTM Process
    - PowerCenter Server Processing
    - System Resources
    - Code Pages and Data Movement Modes
    - Output Files and Cache
- Workflow Schedule
    - Scheduler Options
    - Editing a Workflow
    - Deleting a Workflow
- Session Logs
- Setting Tracing Levels
- Additional Transformations
    - Sequence Generator Transformation
    - Update Strategy Transformation
    - Router Transformation
    - Sorter Transformation
    - Stored Procedure Transformation
    - Union Transformation
    - SQL transformation
    - Reusable Transformations
- Other Tasks
    - Command Task
    - Email Task
    - Assignment Task
    - Worklets
- SQL and Update Override
    - Pre-Post SQL
    - SQL Override and Update Override
    - Update Override

- Indirect Source Files
- Using Multiple Source Files
- Indirect Files
- Mapping Parameter and Mapping Variables
    - Mapping Parameter and Variable Overview
    - Mapping Parameters
    - Mapping Variables
- Dynamic Lookup
    - Static Cache
    - Dynamic Cache
- Slowly changing dimensions
    - Introduction
    - Difference between SCD1 and SCD2
    - Handling Keys
    - Example
    - Numbering Versions
    - Understanding the Mapping
- Commit Points
    - Source-based Commit
    - Target-based Commit
    - User-Defined Commit
- Workflow, Session and Mapping Parameters ·
    - Session Parameters
    - Source File Parameter
    - Target File Parameter
    - Lookup File Parameter
    - Reject File Parameter
- Incremental Aggregation
- Constraint based loading
- Row Error Logging
- Error handling Architecture

IICS

Program Duration: 9 days

- Informatica Cloud Overview
    - Informatica Intelligent Cloud Services (IICS) as an iPaaS solution
    - Informatica Cloud Terminologies
    - Informatica Cloud Architecture
- Runtime Environments and Connections
    - Runtime Environments
    - Secure Agent Architecture
    - Secure Agent Status
    - Elastic Server
    - Secure Agent Services
    - IICS Log Files
    - Connections
    - Connection Types
    - Creating Connections
    - Lab: Installing IICS Secure Agent

    - Lab: Creating a Salesforce connection
    - Lab: Creating a Flat File connection
    - Lab: Creating an Oracle connection
- Synchronization and Data Transfer Task
    - Synchronization Task Overview
    - Synchronization Task – Definition Step
    - Synchronization Task – Source Step
    - Synchronization Task – Target Step
    - Synchronization Task – Data Filters Step
    - Synchronization Task – Field Mapping Step
    - Synchronization Task – Schedule Step
    - Data Transfer Task
    - Activity Monitor
    - Lab: Creating a Synchronization Task
    - Lab: Using Filter, Expression and Lookup in a Synchronization Task
    - Lab: Creating a Data Transfer Task
- Cloud Mapping Designer
    - Cloud Mapping Designer Overview
    - Mapping Designer Terminologies
    - CLAIRE Transformation Recommendations
    - Mapping Data Preview
    - Customizing Mapping Preview
    - Field Rules
    - Best Practices for Creating Mappings
    - Data flow run order
    - Mapping Task
    - Dynamic Schema Change
    - Deploy mapping updates
- Cloud Mapping Designer - Transformations
    - Source Transformation
    - Target Transformation
    - Filter Transformation
    - Joiner Transformation
    - Expression Transformation
    - Lookup Transformation
    - Union Transformation
    - Mapplets
    - SQL Transformation
    - Aggregator Transformation
    - Normalizer Transformation
    - Lab: Using Query in a Mapping
    - Lab: Using Normalizer, Aggregator in a mapping
    - Lab: Using Unconnected Lookup transformation in a mapping
    - Lab: Using Mapplet transformation in a mapping
- Mapping Parameters
    - Parameterization use cases
    - Adding Parameters to a Mapping
    - Creating parameters
    - Parameter Types
    - Using parameter files

    - In-Out Parameter in Mapplet
    - Parameter File Directory
    - Parameter Best Practices
    - Lab: Performing Complete Parameterization
    - Lab: Using Parameter File in a Mapping task
    - Lab: Using In-Out parameters for incremental data loading
- Expression Macro and Dynamic Linking
    - Expression Macro
    - Dynamic Linking
    - Lab: Using Expression Macro in a mapping
    - Lab: Using Dynamic Linking in a mapping
- Replication Task
    - Replication Task Overview
    - Replication Task Features
    - Replication Task: Source and Target Options
    - Other Replication Task Options
    - Resetting the Target Table
    - Generating Non-Unique Index
    - Appendix: Replicating Data to a Flat file
    - Creating a replication task
- Taskflows
    - Taskflow overview
    - Linear Taskflows
    - File Listener Enhancements
    - Taskflow Steps
    - Taskflow Templates
    - Parameters in Taskflow
    - Using REST APIs
    - Invoke a Taskflow through a File Listener
    - Lab: Passing in-out parameters in a Taskflow
    - Lab: Invoking a Taskflow through a File Listener
- Automating and Monitoring Tasks
    - Schedules
    - Schedule Repeat Frequency
    - Schedule Blackout Period
    - Schedule Export
    - Monitoring tasks
    - Email Notifications
    - Event Monitoring
    - Lab: Creating a schedule
- Hierarchical Connectivity
    - Web Service transformation
    - REST V2 Connector
    - Hierarchical Schemas
    - Hierarchy Parser Transformation
    - Hierarchy Builder Transformation
    - Lab: Creating a mapping using a REST V2 connector
    - Lab: Using Web Services transformation in a mapping
    - Lab: Using Hierarchy Parser and Hierarchy Builder Transformation
- Intelligent Structure Model

    - Intelligent Structure Model
    - Intelligent Structure Discovery Process
    - Intelligent Structure Example
    - Refining a Discovered Structure
    - Editing an Intelligent Structure Model
    - Intelligent structure models based on an XSD file
    - Using intelligent structure models in Structure Parser transformation
    - Lab: Creating an Intelligent Structure Model
    - Lab: Using Structure Parser transformation in a mapping
- Advanced Task Settings and Performance Tuning
    - Primary Key Chunking
    - Lookup SQL Override
    - Partitions Overview
    - Types of Partitions
    - Partitioning Rules and Guidelines
    - Pushdown optimization overview
    - Types of pushdown optimization
    - Pushdown optimization user-defined parameters
    - Secure agent groups
    - Shared secure agent groups
    - DTM performance properties
- Exception Handling
    - Types of exceptions
    - User-defined exceptions
    - Non-fatal exceptions
    - Default Field Value Setting
    - Row Error Logging
    - Error handling settings
    - Fatal exceptions
    - Bad files or Reject files
    - Lab: Creating a mapping to handle non-fatal errors